

Bias Mitigation for Harmful Meme Detection using Front-door Adjustment

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Abstract

Although Internet memes are frequently intended to be humorous, they can be deliberately weaponized to disseminate injurious content, exacerbate social cleavages, and incite discrimination. Existing harmful meme detectors predominantly focus on multimodal semantic fusion to capture the nuanced interplay across different modalities. However, they neglect latent data bias, resulting in incorrect predictions based on spurious label-specific features. To remedy this limitation, we propose **CausalHM**, a causally grounded framework that debiases harmful meme detection via front-door adjustment. We first reconstruct a structural causal model in which an unobserved confounder opens a back-door path between memes and harmfulness labels. Then, front-door adjustment blocks the back-door path by introducing reasoning as the mediator between memes and labels. Accordingly, the causal effect of memes on harmfulness labels is decomposed into: the effect of memes on the mediator, quantified by adopting multimodal large language models (MLLMs) with diverse beam search to calculate the probability of each reasoning sequence; and the effect of the mediator on labels, efficiently approximated by the normalised weighted geometric mean (NWGM) method. Finally, a LLM synthesizes post-hoc explanations from the reasoning sequences and predicted labels. Extensive empirical evaluations on four English and Chinese benchmarks demonstrate that CausalHM attains state-of-the-art performance while concurrently furnishing informative explanations.

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

Harmful Meme Detection; Front-door Adjustment; Reasoning; Bias Mitigation; Explainability

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1 Introduction

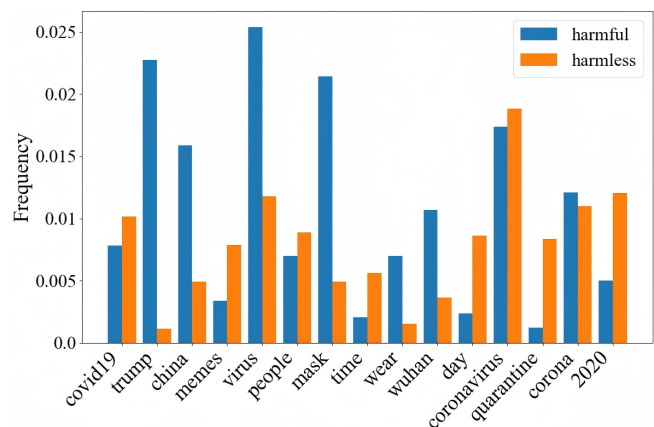


Figure 1: The word frequency distribution of embedded text on different labels (HarmC dataset).

Internet memes — multimodal artefacts that couple an image with a short text — have become a dominant vehicle for public commentary on socio-political events. Their humour and shareability accelerate viral dissemination across social media platforms, yet they can also be weaponized by unscrupulous creators in political and sociocultural conflicts, thereby shifting their impact from entertainment to harm. Harmful memes¹ refer to maliciously exploited memes that use humor or satire as a cover to spread hate speech, discrimination, misleading information, or extremist ideologies, resulting in group stigmatization, online radicalization, and even real-world harm [37]. A paradigmatic case is Donald Trump, whose controversial social-media pronouncements were rapidly repurposed into harmful memes. These memes not only inflicted reputational harm on Trump personally, but also amplified stigmatizing discourse that could exacerbate international confrontation between the United States and China, particularly through the use of terms such as “Trump virus” and “China Virus”. Therefore, efforts to protect users from toxic memes necessitate the development of automatic detectors capable of identifying harmful memes.

¹Disclaimer: This paper contains content that may be disturbing to some readers.

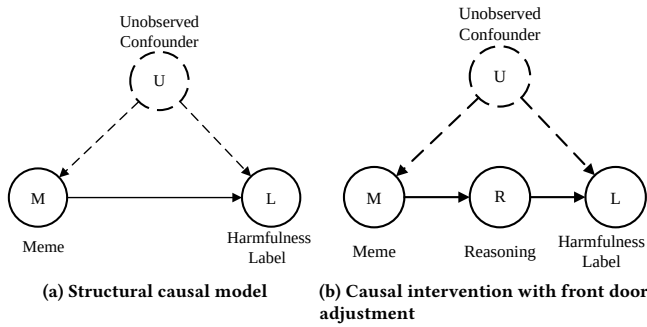


Figure 2: Structural causal model for harmful meme detection.

Identifying harmful memes is inherently challenging due to the abstruse nature of memes. Earlier work [32] leveraged pre-trained visual and textual unimodal detectors like BERT [9] and Faster R-CNN [34] to learn representations in isolation for prediction. Nevertheless, the strategic amalgamation of images and text prioritizes methods comprehensively analyzing the nuanced interplay across different modalities over isolated analysis of each modality. A considerable number of studies [15, 17, 18, 36] are dedicated to capturing multimodal features from images, text, and implicit context to detect the harmfulness of memes. MOMENTA [33] aligned local and global unimodal features with background context. Pro-Cap [2] introduced a probing-based captioning technique by leveraging the frozen pre-trained vision-language model (PVLm) to yield descriptive image captions. Owing to the powerful semantic understanding capabilities of multimodal large language models (MLLMs) [46], MLLMs were employed to enrich the background contextual information. ExplainHM [21] elicited MLLMs in a debate manner to generate dialectical rationales from both harmful and harmless standpoints which not only facilitated multimodal fusion but also served as explanations behind decisions.

Despite their notable success in discerning harmful memes, these methods remain vulnerable to unexpected data bias that may mislead models to explore spurious correlations between meme patterns and labels. Due to the human-centric nature of data creation and collection, human annotators and data curators inevitably embed various implicit or explicit biases into datasets, including entity bias [51], psycholinguistic bias [5], *etc.* Further compounding this challenge, data bias exhibits high diversity and sophistication, thereby rendering manual enumeration and identification of all potential bias sources a computationally and cognitively intractable task. Take the HarmC dataset [32] as example, as showcased in Figure 1, word frequency statistics uncover that terms such as "Trump", "mask", "China" and "Wuhan" exhibit strong statistical correlation with the label "HARMFUL" while "quarantine", "day" and "memes" are disproportionately associated with the label "HARMLESS". As a consequence, models learn spurious label-specific features from these biases as shortcuts rather than understanding the semantic rationales when training, which unavoidably undermines their ability to generalize to unseen test data where such biases are eliminated.

Recently, causal inference [31] has emerged as a preferred paradigm for developing debiased models through causal intervention or counterfactual reasoning across various tasks [25, 44, 47]. As

illustrated in Figure 2(a), conventional approaches directly estimate the total causal effect $P(L | M)$ of input memes (M) on output labels (L) while overlooking the unobserved confounder (U). From a causal inference standpoint, the presence of U introduces a back-door path [31] $M \leftarrow U \rightarrow L$, thereby mixing the genuine causal effect of M on L with spurious correlations. However, back-door adjustment is infeasible here because the unobserved confounder U absorbs various biases which cannot be exhaustively enumerated and identified.

To remedy this issue, we propose **CausalHM**², a novel debiasing framework that adopts the front-door adjustment to disentangle causal signals from bias, enabling robust and causally grounded predictions for harmful meme detection. As formalized in Figure 2(b), *reasoning* (R), independent of the unobserved confounder U , is introduced as the mediator that bridges memes (M) and harmfulness labels (L). The causal effect of M on L is then decomposed into two components: the causal effect of M on R and the causal effect of R on L . Specifically, to quantify the causal effect between M and R , we draw inspiration from [25] and instantiate R using reasoning sequences generated by multimodal large language models (MLLMs) with diverse beam search [38]. To calculate the causal effect between R and L , we apply the normalized weighted geometric mean (NWGM) [41, 47, 48] approximation to yield unbiased outcomes based on the generated reasoning sequences. Furthermore, to render the model's predictions transparent, we prompt a LLM to synthesize post-hoc explanations from reasoning and the predicted harmfulness labels.

Our principal contributions are four-fold:

- (1) To the best of our knowledge, we are the first to debias harmful meme detection from the perspective of casual inference.
- (2) We propose **CausalHM** that performs causal intervention with front-door adjustment by introducing *reasoning* between memes and harmfulness labels as the mediator. The mediator is instantiated by reasoning sequences generated by MLLMs with diverse beam search.
- (3) We leverage a LLM to produce informative explanations that integrate reasoning with the final detection decision, thereby surfacing latent semantic cues embedded in the multimodal meme content.
- (4) Extensive experiments on four English and Chinese datasets demonstrate the superiority of our proposed **CausalHM** over other non-debiased competitors in both model generalization and interpretability.

2 Related Work

Harmful Meme Detection. Early efforts in harmful meme detection primarily relied on unimodal representation learning, applying pre-trained encoders like BERT [9] and Faster R-CNN [34] to either the textual or visual component of memes in isolation. [32] revealed that unimodal detectors exhibited limited performance since the strategic interplay between image and text was extraordinarily complex. Mainstream studies have converged on integrating textual, visual, and even contextual information to capture the nuanced interplay across modalities. MaskPrompt [3] devised a multimodal prompt and applied a pre-trained language model to generate the

²<https://github.com/critivitycheng/CausalHM>

harmfulness label. RGCL [30] learned a hatefulness-aware vision and language joint embedding space via contrastive learning. With the advent of LLMs, numerous LLM-enhanced frameworks have been designed to enrich background context, or even enable few-shot or zero-shot detection [12, 23, 29, 35]. MR.HARM [22] first distilled multimodal reasoning knowledge from MLLMs and then fine-tuned a smaller language model to aggregate multimodal features. HMGUARD [52] integrated adaptive prompting with chain-of-thought (CoT), leveraging MLLMs to encode the multimodal features. MIND [26] organized multi-LLMs as an autonomous agent to conduct zero-shot harmful meme detection. Recently, some researchers shifted their attention from detection accuracy to interpretability. ExplainHM [21] elicited MLLMs in a debate manner to generate dialectical rationales as explanations. Gu et al. [10] introduced the MemeMind dataset containing CoT annotations and devised MemeGuard which effectively integrated multimodal information with reasoning. Nowadays, Chinese harmful meme detection has also received attention: the ToxiCN MM corpus was released together with MKE [28], a knowledge-enhanced framework that injects external Chinese commonsense; and the CHMEMES corpus was released together with SCARE [11], a semantic contrastive alignment framework that enables a comprehensive representation of both cross-modal and intra-modal information.

Nevertheless, the aforementioned approaches overlook hidden data bias, thereby learning spurious label-specific features that impair generalization when such biases are absent at test time. In this study, we mitigate these unobserved confounders from the perspective of causal inference.

Causal Inference. More recent attention has been paid to developing unbiased models through causal inference [31, 50]. Back-door adjustment and front-door adjustment are two widely-applied causal intervention tools, estimating the causal effect between treatment and outcome via do-calculus. With the aid of back-door adjustment, CLIMB [13] removed image–text matching bias; CCD [5] mitigated psycholinguistic bias and resolved image-specific bias through counterfactual reasoning; and SuCI [44] eliminated subject bias in multimodal language understanding. Unfortunately, back-door adjustment collapses when confounders are unobservable [42]. Front-door adjustment therefore offers an appealing alternative for estimating causal effects in the presence of hidden confounders. Assisted by front-door adjustment, Causal Walk [48] inserted reasoning paths as the mediator between claim-evidence graph and veracity label; Causal Prompting [47] employed LLM-generated chains of thought to mediate between prompts and answers; and FABE [25] mapped input text to semantically equivalent variables to defend against back-door attacks in LLMs. Inspired by these successes, we intend to debias harmful meme detection with front-door adjustment by introducing *reasoning* as the mediator between input memes and output labels.

3 Preliminaries

3.1 Structural Causal Model and Causal Effect

Given a meme $m = (I, T) \in M$, where I denotes the visual content and T the associated textual sequence, the objective of harmful meme detection is to automatically infer its corresponding harmfulness label $L \in \{\text{harmful}, \text{harmless}\}$. In this study, we seek to mitigate hidden data bias with the aid of causal inference. As shown

in Figure 2(a), the detection of harmful memes is represented as a structural causal model (SCM) [31] $G = \{V, E\}$, where V is the set of variables and E the set of directed edges encoding causal dependencies. The directed edge $M \rightarrow L$ denotes the direct causal influence of input memes M on output labels L . However, due to the human-centric nature of data creation and collection, human annotators and data curators inevitably embed various implicit or explicit biases into datasets. As a consequence, we introduce the confounder U into the SCM which opens a back-door path $M \leftarrow U \rightarrow L$ between the treatment M and the outcome L .

Conventional detectors typically frame harmful meme detection as a direct estimation of the statistical association $P(L | M)$, relying heavily on spurious label-specific features learned from the biased data. From the perspective of causal inference, the total effect of M on L is the aggregation of the front-door path $M \rightarrow L$ and the back-door path $M \leftarrow U \rightarrow L$. Therefore, detectors that merely fit $P(L | M)$ fail to distinguish between causal signals and spurious correlations, leading to performance degradation on unseen test data where such biases are eliminated. To address this issue, we need to isolate the true causal effect of M on L free from the confounding influence of U , which requires blocking the back-door path $M \leftarrow U \rightarrow L$. Causal intervention can be applied to estimating the true causal effect of M on L by using the do-calculus [31] $P(L | do(M))$:

$$P(L | do(M)) = \sum_u P(L | M, u)P(u) \quad (1)$$

where $u \in U$ means to be any latent bias that simultaneously influences both the treatment M and the outcome L .

3.2 Causal Intervention with Front-door Adjustment

The backdoor adjustment is infeasible to compute Equation 1 because the unobserved confounder U absorbs various biases which cannot be exhaustively enumerated and identified. Instead, we resort to the front-door adjustment which yields a bias-free estimate of the causal effect of M on L without necessitating access to U . The pivotal requirement for front-door adjustment is the identification of a mediator that fully mediates the causal effect of the treatment on the outcome, ensuring that every direct causal path from the treatment to the outcome is channeled exclusively through the mediator. As shown in Figure 2(b), we introduce reasoning as the mediator variable R between memes M and harmfulness labels L . The true causal effect of M on L can then be formulated as:

$$P(L | do(M)) = \sum_{r \in R} P(L | do(r))P(r | do(M)) \quad (2)$$

Here, each $r \in R$ denotes a reasoning instance associated with M . Accordingly, the true causal effect of M on L is decomposed into two successive transmissions: (i) $P(r | do(M))$, capturing the causal influence exerted by M on the mediator R , and (ii) $P(L | do(r))$, quantifying the effect of R on the outcome L .

Estimating $P(r | do(M))$. Owing to the collider structure $U \rightarrow L \leftarrow R$, the back-door path $M \leftarrow U \rightarrow L \leftarrow R$ is d-separated and thus automatically blocked, ensuring that the causal effect of R on L remains unconfounded. Thus, we have

$$P(r | do(M)) = P(r | M) \quad (3)$$

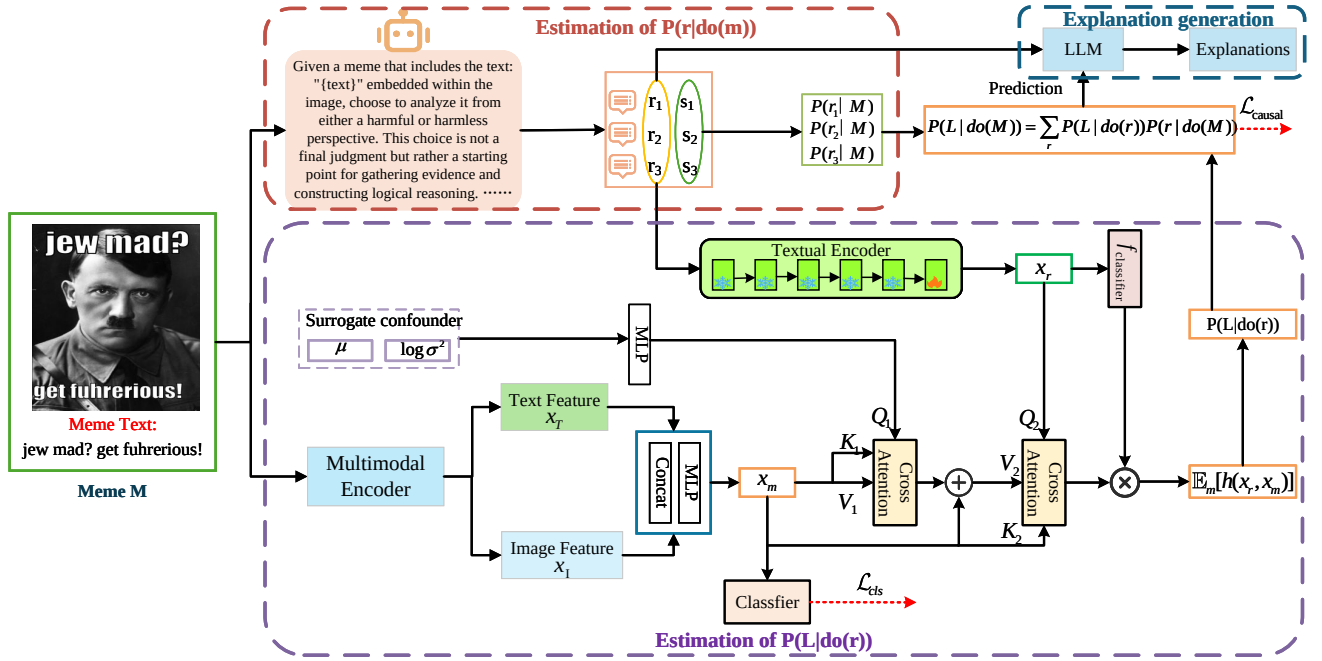


Figure 3: The framework of our proposed CausalHM. First, we estimate $P(r | do(M))$ using MLLMs with diversity beam search. Next, we apply the NWGM to approximate an unbiased estimate of $P(L | do(r))$. Finally, a LLM synthesizes explanations based on reasoning and predictions.

Estimating $P(L | do(r))$. Conditioning on M blocks the back-door path $R \leftarrow M \leftarrow U \rightarrow L$, thereby satisfying the back-door criterion. Hence, the causal effect of R on L can be estimated as

$$P(L | do(r)) = \sum_m P(L | r, m) P(m) \quad (4)$$

4 Methodology

As delineated in Figure 3, CausalHM integrates three synergistic modules: (i) Estimating of $P(r | do(M))$ (4.1), (ii) Estimation of $P(L | do(r))$ (4.2) and (iii) Explanation Generation (4.3). Firstly, we quantify $P(r | do(M))$ by leveraging MLLMs to generate reasoning together with sequence scores from either a harmful or harmless perspective. Secondly, the NWGM is used to approximate an unbiased estimate of $P(L | do(r))$. Thirdly, we employ diverse beam search to estimate $P(L | do(M))$ and optimize the model with our well-designed objective function (4.4). Finally, a LLM yields informative explanations based on reasoning and predictions.

4.1 Estimation of $P(r | do(M))$

As formalized in Equation (3), the interventional distribution $P(r | do(M))$ is equivalently expressed as $P(r | M)$, thereby enabling direct estimation from the empirical data. The remaining challenge therefore centres on the sampling of the front-door variable R . Contemporary MLLMs have achieved substantial advances in comprehending multimodal semantics, extending beyond surface-level text and image cues to encompass implicit knowledge such as common-sense knowledge and cultural context. Following [25], we employ MLLMs to sample the front-door variable.

Specifically, given a meme sample $m = \{I, T\}$, we curate a prompt template p^r that embeds the textual component T and is subsequently fed to a vision-language model (e.g., LLaVA [24]). Based on this prompt, the MLLM generates one piece of reasoning r articulated from either a harmful or harmless standpoint. The prompt template p^r is designed as follows:

"Given a meme that includes the text: $[T]$ embedded within the image, choose to analyze it from either a harmful or harmless perspective. This choice is not a final judgment but rather a starting point for gathering evidence and constructing logical reasoning. Examine how the visual and textual elements interact within broader cultural, social, and psychological frameworks. Assess whether the meme relies on stereotypes, references sensitive historical or political events, or generates strong emotional responses, and uncover how these clues might shape the viewer's interpretation. Under the selected perspective, pinpoint specific features within the meme that underpin the reasoning. The objective is to practice formulating structured reasoning by compiling relevant evidence and explaining how it aligns with the chosen perspective, rather than issuing a conclusive judgment."

Front-door adjustment necessitates a portfolio of high-probability yet distinct reasoning sequences. [25] used the standard beam search to sample multiple pieces of reasoning as the front-door variables, resulting in insufficient diversity among reasoning sequences. Instead, we deploy the diverse beam search [38] to sample multiple distinct reasoning sequences. Diverse beam search augments vanilla beam search by explicitly penalizing redundancy. Candidate sequences are partitioned into groups at each decoding step, and a group-wise diversity penalty suppresses intra-group similarity, thereby yielding a heterogeneous set of high-probability outputs.

Formally, we leverage the MLLM with the diverse beam search to produce: (i) a set of B reasoning sequences, denoted as $\{r_i\}_{i=1}^B$, and (ii) an associated set of B sequence scores, denoted as $\{s_\theta(r_i)\}_{i=1}^B$. B is the beam width. The scoring function $s_\theta(r_i)$ is defined as:

$$s_\theta(r_i) = \frac{1}{|r_i|} \log F(r_i | p^r; \theta) \quad (5)$$

where $F(r_i | p^r; \theta)$ represents the probability that given the prompt p_r the vision LLM $F(\cdot | \cdot; \theta)$ generate sequence r_i , and $|r_i|$ denotes the length of sequence r_i , preventing longer sequences from being unduly penalized.

To approximate the probability distribution over these sequences, we apply the softmax function to the scores, yielding the probability of each reasoning r_i as:

$$P(r_i | M) \approx \frac{\exp(s_\theta(r_i))}{\sum_{k=1}^B \exp(s_\theta(r_k))} \quad (6)$$

4.2 Estimation of $P(L | do(r))$

As formalized in Equation (4), the back-door adjustment transformed the interventional distribution $P(L | do(r))$ to $\sum_m P(L | r, m) P(m)$. We first derive meme representations and reasoning representations through the **Multimodal Meme Encoding** and **Reasoning Encoding**, respectively. Then, in **Reasoning Deconfounding**, we apply the NWGM approximation to compute the causal effect efficiently. At last, in **Reasoning Classifying**, we compute the label probability distribution based on the reasoning representation after causal intervention.

Multimodal Meme Encoding. Visual and textual features of a meme m are jointly extracted via a multimodal image-text encoder [43]:

$$x_I, x_T = \text{MultimodalEncoder}(I, T) \quad (7)$$

$$x_m = \text{MLP}([x_I, x_T]) \quad (8)$$

The encoder emits visual and textual embeddings (x_I and x_T), which are concatenated and forwarded through a multi-layer perceptron (MLP) to produce the fused multimodal representation x_m .

Reasoning Encoding. The reasoning r , generated by vision LLMs, is encoded using the pre-trained BERT model [9]:

$$x_r = \text{BERT}(r) \quad (9)$$

where x_r represents the contextualized embedding of reasoning r .

Reasoning Deconfounding. Given the meme representation x_m and reasoning representation x_r , the interventional distribution $P(L | do(r))$ is transformed to expectation as follows:

$$P(L | do(r)) = \sum_{m \in M} P(x_m) P(L | x_r, x_m) = \mathbb{E}_m[P(L | x_r, x_m)] \quad (10)$$

wherein $P(L | x_r, x_m)$ signifies the predicted probability of label l estimated by the classifier $f_{\text{classifier}}$:

$$P(l | x_r, x_m) = f_{\text{classifier}}(x_r, x_m) = \text{softmax}(h(x_r, x_m)) \quad (11)$$

where h is a feature fusion layer integrating x_r with x_m .

As the expectation $\mathbb{E}_m$ is analytically intractable, we approximate it efficiently by means of the NWGM method [41, 47, 48]:

$$\mathbb{E}_m[\text{softmax}(h(x_r, x_m))] \approx \text{softmax}(\mathbb{E}_m[h(x_r, x_m)]) \quad (12)$$

Following established prior work [48, 49], we realize the fusion function h via a dedicated neural network layer parameterized as:

$$h(x_r, x_m) = W_r x_r + \alpha W_m x_m \quad (13)$$

where W_r and W_m denote learnable weight matrices, and α is a weight coefficient for intervention. With this parameterization, we instantiate this expectation as:

$$\mathbb{E}_m[h(x_r, x_m)] = W_r x_r + \alpha W_m \mathbb{E}_m[x_m] \quad (14)$$

Given the high dimensionality of the meme space and the presence of latent confounders, obtaining an exact estimate of $\mathbb{E}_m[x_m]$ is challenging. To address this issue, we employ the deconfounder algorithm [6, 40] to infer surrogate confounders. Specifically, we introduce a latent variable $Z \sim \mathcal{N}(\mu, \sigma^2)$ as the distribution of surrogate confounders:

$$\sigma = \exp(\frac{1}{2} \log \sigma^2) \quad (15)$$

$$\epsilon \sim \mathcal{N}(0, I_{n \times d}) \quad (16)$$

$$z_{ij} = \mu_{ij} + \exp(\frac{1}{2} \log \sigma_{ij}^2) \epsilon_{ij} \quad (17)$$

where $\mu, \sigma^2 \in \mathbb{R}^{n \times d}$ are learnable parameters initialized to zeros. n is the number of tokens in x_m and d is the feature dimension. The sampled latent variable z encodes confounding-aware feature representations.

To estimate $\mathbb{E}_m[x_m]$, we devise two successive cross-attention layers. In the first layer, z serves as the query while the meme representation x_m acts as both the key and value. The attention mechanism is thus instantiated as:

$$z' = \text{MLP}(z) \quad (18)$$

$$Q_1 = W_q^1 z', \quad K_1 = W_k^1 x_m, \quad V_1 = W_v^1 x_m \quad (19)$$

$$z'' = \text{softmax}(Q_1^T K_1) V_1 \quad (20)$$

$$x'_m = x_m + z' \quad (21)$$

The resulting vector x'_m integrates confounding-aware contextual signals into the meme representation. Then, in the second layer, the reasoning representation x_r serves as the query while the original meme representation x_m and the confounding-aware meme representation x'_m act as the key and value respectively. The attention mechanism is thus instantiated as:

$$Q_2 = W_q^2 x_r, \quad K_2 = W_k^2 x_m, \quad V_2 = W_v^2 x'_m \quad (22)$$

$$z''' = \text{softmax}(Q_2^T K_2) V_2 \quad (23)$$

Reasoning classifying. We first use the classifier function $f_{\text{classifier}}$ to compute the label probability distribution based on the reasoning representation x_r . Then $\mathbb{E}_m[x_m]$ is formalized as:

$$P(l | x_r) = f_{\text{classifier}}(x_r) \quad (24)$$

$$\mathbb{E}_m[x_m] = P(l | x_r) \cdot z''' \quad (25)$$

To sum up, with the help of the NWGM approximation, the interventional distribution $P(L | do(r))$ is deduced as:

$$P(L | do(r)) = \text{softmax}(W_r x_r + \alpha W_m \mathbb{E}_m[x_m]) \quad (26)$$

4.3 Explanation Generation

To overcome the opacity of black-box detectors, this section synthesizes human-readable textual explanations based on reasoning R and prediction results, surfacing the implicit meaning embedded in memes. Specifically, given a meme sample $m = \{I, T\}$, we construct a prompt template p^e that integrates the meme text T , its corresponding reasoning $R = \{r_1, r_2, \dots, r_B\}$ generated by MLLMs, and the final predicted harmfulness label l_{pred} . The prompt template p^e is structured as follows:

You are given a meme containing the embedded text: $[T]$. Below are $[B]$ pieces of reasoning from either a harmful or harmless viewpoint: $[R]$. Your task is two-fold: (i) extract the reasons that support the classification of this meme as $[l_{\text{pred}}]$ from the reasoning; (ii) provide a clear and structured explanation that justifies this prediction on the basis of the selected reasons.

Based on this prompt template, the vision LLM distills evidence from multiple reasoning sequences, aligns the selected arguments with the predicted label l_{pred} , and generates an explanation e that is simultaneously human-readable and information-rich.

4.4 Loss function

To ensure the model encodes causal relations while learning discriminative representations, we devise a joint objective comprising the **causal loss** and the **classification loss**.

Causal Loss. The front-door adjustment introduces reasoning as the mediator R , yielding a bias-free estimate of the causal effect of M on L without necessitating access to the unobserved confounder U . In accordance with Equations 2, 6 and 26, we can implement front-door adjustment as:

$$l_{\text{causal}} = P(L \mid \text{do}(M)) \quad (27)$$

$$= \sum_{r \in R} \frac{\exp(s_\theta(r))}{\sum_{k=1}^B \exp(s_\theta(r_k))} \text{softmax}(W_r x_r + \alpha W_m \mathbb{E}_m[x_m]) \quad (28)$$

$$= \mathbb{E}_{r \in R} [\text{softmax}(W_r x_r + \alpha W_m \mathbb{E}_m[x_m])] \quad (29)$$

Specifically, the binary cross-entropy loss is adopted for training:

$$\mathcal{L}_{\text{causal}} = -l_{\text{gold}}^i \log l_{\text{causal}}^i - (1 - l_{\text{gold}}^i) \log(1 - l_{\text{causal}}^i) \quad (30)$$

where l_{gold}^i denotes the ground-truth label, and l_{causal}^i is the predicted probability based on front-door adjustment.

Classification Loss. To further enhance the discriminative capacity of multimodal representations, we introduce an auxiliary classification loss formulated as binary cross-entropy:

$$l_{\text{cls}} = f_{\text{classifier}}(x_m) \quad (31)$$

$$\mathcal{L}_{\text{cls}} = -l_{\text{gold}}^i \log l_{\text{cls}}^i - (1 - l_{\text{gold}}^i) \log(1 - l_{\text{cls}}^i) \quad (32)$$

where l_{cls}^i represents the predicted probability based solely on the multimodal features x_m .

Joint Loss Function. To balance these complementary objectives, the overall optimization criterion is formulated as a weighted aggregation of the causal and classification losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{causal}} + \lambda \mathcal{L}_{\text{cls}} \quad (33)$$

where the trade-off hyper-parameter λ calibrates the relative influence of the causal and classification losses.

5 EXPERIMENTS

5.1 Experimental Setup

Datasets. We assess CausalHM on four real-world meme benchmarks: HarmC [32], HarmP [33], FHM [17], and ToxiCN-MM [28]. As for HarmC and HarmP, originally annotated with three labels: *very harmful*, *somewhat harmful*, and *not harmful*, we collapse *very harmful* and *somewhat harmful* into a single *harmful* class to align with prior binarized evaluations [2, 3, 21, 33]. As for FHM, we manually annotate the unlabelled test split [3]. ToxiCN-MM³, the first Chinese harmful meme dataset, consists of 12,000 samples. To preserve a standard train-validation-test split, we further partition the official training set into 8,400 training and 1,200 validation instances while keeping the original test set intact.

Baselines. CausalHM is evaluated against 10 competitors: 1) **Text BERT** [1, 8]; 2) **Image-Region** [34]; 3) **Late Fusion** [32]; 4) **MMBT** [16]; 5) **VisualBERT COCO** [20]; 6) **ViLBERT CC** [27]; 7) **MOMENTA** [33]; 8) **MaskPrompt** [3]; 9) **Pro-Cap** [2]; 10) **ExplainHM** [21]. In line with prior work [2, 3, 21, 33], accuracy and macro-F1 score are reported as the evaluation metrics.

We append the data statistics, baseline descriptions, and implementation details in Appendices A, B, and C, respectively.

5.2 Performance on Harmful Meme Detection

Table 1 summarizes the harmful meme detection results on the four datasets, revealing the following salient findings:

1) Expectedly, our proposed CausalHM consistently delivers the state-of-the-art performance across all metrics and datasets, outperforming the strongest non-debiased baseline (ExplainHM) by 2.08%, 0.82%, 2.64%, and 1.81% in Macro-F1 on Harm-C, Harm-P, FHM, and ToxiCN-MM, respectively. This confirms our hypothesis that non-debiased methods suffer heavily from hidden data bias, and demonstrates the efficacy of our debiasing framework with front-door adjustment in mitigating such unobserved bias.

2) Compared to ExplainHM, CausalHM yields substantially larger improvements on Harm-C, FHM, and ToxiCN-MM, while achieves comparatively modest gains on Harm-P. This indicates that Harm-C, FHM, and ToxiCN-MM are more susceptible to hidden data bias than Harm-P, and ExplainHM learned spurious label-specific features during training which didn't generalize well to the testing set. The comparatively modest gains on Harm-P may be attributed to the political domain's propensity for memes that repeatedly target identical figures or policies, thereby inducing spurious correlations that remain stable across training and test splits and consequently reduce the apparent benefit of causal debiasing.

3) Multimodal detectors demonstrate superior performance compared to unimodal counterparts, underscoring the critical importance of modeling fine-grained cross-modal interactions.

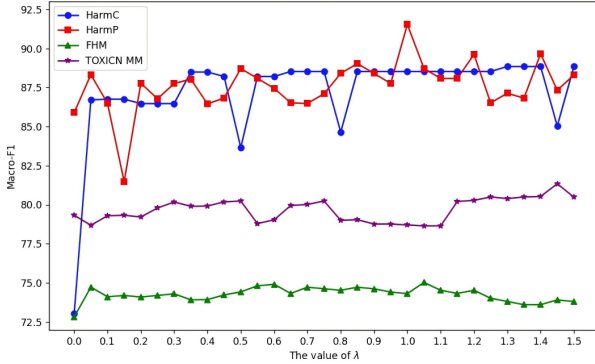
4) Among multimodal detectors, those that exploit LLMs to enrich background context systematically beat other baselines, suggesting the promising potential of LLMs for surfacing subtle harmful cues. For example, ExplainHM outperforms MMBT by 19.29%, 10.49%, 9.12%, and 2.75% in Macro-F1 on Harm-C, Harm-P, FHM, and ToxiCN-MM, respectively.

³<https://github.com/DUT-lujunyu/ToxiCN-MM>

Table 1: Harmful meme detection results on four datasets. The numbers marked with $\diamond$ are taken from [21]. We mark the best and second-best results in bold and underlined, respectively.

Dataset	HarmC		HarmP		FHM		TOXICN MM	
Model	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1
Text BERT [1, 8]	70.17 $\diamond$	66.25 $\diamond$	80.12 $\diamond$	78.35 $\diamond$	63.48	64.80	79.92	76.61
Image-Region [34]	68.74 $\diamond$	62.97 $\diamond$	73.14 $\diamond$	74.27 $\diamond$	51.20	49.88	69.25	63.93
Late Fusion [32]	73.24 $\diamond$	70.25 $\diamond$	78.26 $\diamond$	76.50 $\diamond$	64.20	63.48	81.00	76.69
MMBT [16]	73.48 $\diamond$	67.12 $\diamond$	82.54 $\diamond$	80.23 $\diamond$	64.40	63.26	80.63	76.78
VisualBERT COCO [20]	81.36 $\diamond$	80.13 $\diamond$	86.07 $\diamond$	86.07 $\diamond$	65.00	64.89	73.25	68.15
ViLBERT CC [27]	78.70 $\diamond$	77.09 $\diamond$	87.25 $\diamond$	86.01 $\diamond$	65.70	63.67	73.79	69.09
MOMENTA [33]	83.82 $\diamond$	82.84 $\diamond$	88.24 $\diamond$	88.26 $\diamond$	66.60	64.72	76.13	69.97
MaskPrompt [3]	84.47 $\diamond$	81.51 $\diamond$	88.17 $\diamond$	87.09 $\diamond$	70.40	70.00	81.79	79.00
Pro-Cap [2]	85.01 $\diamond$	83.17 $\diamond$	89.32 $\diamond$	87.91 $\diamond$	71.95	71.30	81.62	79.34
ExplainHM [21]	87.00 $\diamond$	86.41 $\diamond$	90.73 $\diamond$	90.72 $\diamond$	72.90	72.38	82.31	79.53
CasualHM	89.27	88.49	91.54	91.54	75.02	75.02	83.92	81.34
w/o intervention	85.03	83.81	84.01	84.01	68.20	68.53	81.13	77.48

5) Among unimodal detectors, Image-Region lags behind Text BERT across all metrics and datasets, indicating that the predominance of textual cues over visual ones in determining meme harmfulness. Identical visual elements can assume divergent harmfulness contingent on the accompanying text, making purely visual methods more susceptible to bias.

**Figure 4: Impact of λ on F1-score across four datasets.**

5.3 Ablation Study

To investigate the contribution of causal intervention, we ablate the front-door adjustment part (denoted w/o intervention) by directly concatenating the multimodal meme encoding x_m with the reasoning encoding x_r for classification. It can be clearly seen from Table 1 that excising the intervention part leads to a marked degradation: Macro-F1 drops from 88.49% to 83.81% on HarmC, from 91.54% to 84.01% on HarmP, from 75.02% to 68.20% on FHM and from 81.34% to 77.48% on ToxicN-MM. These reductions underscore the indispensable role of front-door adjustment in bias mitigation.

5.4 Sensitivity analysis

The hyper-parameter λ governs the relative contribution of the causal loss and the classification loss. We performed a grid search over $\lambda \in [0.0, 1.5]$ on the test splits of all four datasets. Figure 4 shows that Macro-F1 curves first rise with mild oscillations and

then fall with mild oscillations. This phenomenon illustrates the empirical trade-off imposed by front-door adjustment: if $\lambda < 1$, the optimization is dominated by the causal loss, yielding poorly discriminative multimodal features and becoming easier to overfit especially when λ is too small; otherwise, the classification loss dominates, and the classifier is optimized almost exclusively for empirical risk when λ is too big, exposing model to hidden data bias. Peak Macro-F1s are attained at $\lambda = 1.3, 1.0, 1.05$, and 1.45 on HarmC, HarmP, FHM, and TOXIC-MM, respectively.

5.5 Evaluation of Explainability

Automatic Evaluation. Conventional metrics fall short in evaluating the quality of LLM-generated explanations [4]. Prior research has shown that ChatGPT-based assessors align closely with expert human judges [7, 14]. Following established studies [21, 39], we employ GPT-5-mini to evaluate each explanation along four dimensions: 1) Informativeness reflects extent of novel background or contextual knowledge introduced; 2) Readability denotes grammatical correctness and structural clarity; 3) Soundness evaluates logical consistency and validity of the stated arguments; 4) Persuasiveness means convincingness and compellingness of the narrative. Quantification is performed on a five-point Likert scale, wherein 1 denotes the lowest and 5 the highest achievable quality.

Table 2 reports the average ratings assigned by GPT-5-mini on HarmC test set and part of TOXICN-MM test set (*i.e.* randomly selecting 1,000 instances) for CausalHM versus ExplainHM. The results reveal two important findings.

1) CausalHM surpasses ExplainHM regarding Informativeness, Soundness, and Persuasiveness on both datasets. This advantage stems from their distinct generation pipelines: ExplainHM merely use the rationale of the “winning” LLM in a multimodal debate, whereas CausalHM aggregates reasoning from both harmful and harmless viewpoints and distills the evidence along with the predicted label. Therefore, CausalHM yields explanations that are simultaneously more comprehensive, logically coherent, persuasive, and information-rich.

2) In terms of readability, CausalHM scores slightly lower than ExplainHM on both datasets. This gap may be attributed to GPT-5-mini’s preference for brevity. In the English dataset, ExplainHM

produces an average of 125.38 words per explanation, whereas CausalHM averages 253.15 words. Similarly, in the Chinese dataset, ExplainHM averages 144.31 words per explanation, while CausalHM averages 338.58 words. These results suggest that the richer level of detail provided by CausalHM is penalized under a conciseness-oriented evaluation metric. Meanwhile, longer mean explanation length correlates positively with informativeness.

Table 2: Automatic GPT-5-mini evaluation of the explanation quality on harmful memes in HarmC and TOXICN MM datasets.

Explanations	HarmC		TOXICN MM	
	ExplainHM	CausalHM	ExplainHM	CausalHM
Informativeness	2.96	3.27	3.48	3.68
Readability	4.36	3.71	4.48	3.89
Soundness	2.97	3.27	3.44	3.71
Persuasiveness	3.06	3.28	3.42	3.73

Human Evaluation. To alleviate the burden of manual assessment, we conduct a human evaluation on 100 randomly selected samples from each test split (HarmC and TOXICN-MM). 10 professional linguistic annotators are then asked to rate the explanations using metrics employed in automatic evaluation.

The results of human evaluation are presented in Table 3, from which we derive two principal findings:

1) Consistent with GPT-5-mini’s assessment, the explanation scores of CausalHM exceeds ExplainHM across Informativeness, Soundness, and Persuasiveness. This infers that our method yields explanations of demonstrably superior quality.

2) CausalHM also marginally surpasses ExplainHM in Readability. Its superior logical coherence and persuasive force compensate for the slightly greater length, yielding explanations that raters judge easier to follow.

Table 3: Human evaluation of the explanation quality in HarmC and TOXICN-MM.

Explanations	HarmC		TOXICN MM	
	ExplainHM	CausalHM	ExplainHM	CausalHM
Informativeness	3.67	3.86	3.60	4.07
Readability	3.90	3.96	3.83	3.95
Soundness	3.59	3.91	3.51	4.01
Persuasiveness	3.63	3.89	3.56	4.03

5.6 Case Study

Does CausalHM take effect on mitigating bias? Figure 1 finds that “Donald Trump” and “2020” are strongly correlated with “harmful” and “harmless”, respectively. As showcased in Figure 5, ExplainHM exploits these spurious correlations and mispredicts, whereas our CausalHM suppresses the shortcuts and yields correct decisions — evidence that front-door adjustment effectively removes the embedded bias.

Does LLM-generated reasoning help to discern harmfulness? From the explanations we observe that the LLM’ parametric knowledge accurately interprets multimodal content embedded in memes. In Figure 5(a), the character Trump in the image is recognized and described as a well-known figure with controversial statements and

actions. Figure 5(b) is illustrated to be a harmful meme referencing the COVID-19 conspiracy theory. Treating such reasoning as the mediator variable, enabling CausalHM to base its decision on genuine rationales.

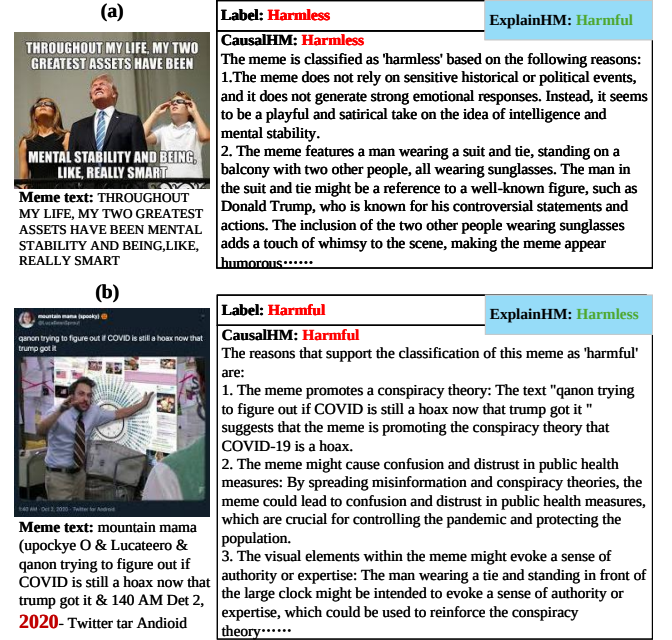


Figure 5: Case study. In contrast to ExplainHM, CausalHM correctly detects two biased examples in HarmC.

6 Conclusion and Future Work

We proposed CausalHM that debiases harmful meme detection with front-door adjustment by introducing reasoning between memes and harmfulness labels as the mediator. The true causal effect of M on L is accordingly decomposed into two components by the mediator. First, the effect of memes on the mediator was transformed to the probability of each reasoning sequence, estimated by MLLMs with diverse beam search. Second, the effect of the mediator on labels was efficiently approximated via the NWGM method. Moreover, a LLM was leveraged to yield informative explanations based on reasoning and predictions. Extensive experiments on four datasets showed CausalHM outperformed 10 competitors in both generalizability and interpretability. Despite the success, we intend to investigate a new explanation generation mechanism that enables explanations to reflect the debiasing effect on the model.

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A DATASETS AND METRICS

Summary statistics for the four datasets are presented in Table 4.

Table 4: Dataset statistics for harmful and harmless instances.

Datasets	Train		Val		Test	
	#harmful	#harmless	#harmful	#harmless	#harmful	#harmless
Harm-C	1064	1949	61	116	124	230
Harm-P	1486	1534	86	91	173	182
FHM	3050	5450	250	250	500	500
TOXICN MM	2700	5700	375	825	765	1635

B BASELINES

Ten SOTA methods were chosen to compete our propose CausalHM for harmful meme detection: 1) **Text BERT** [1, 8], a text-only approach that leverages BERT for classification. 2) **Image-Region** [34], a vision-only model that extracts image features using Faster R-CNN with ResNet-152. 3) **Late Fusion** [32], a multimodal method that combines prediction scores from BERT and ResNet-152. 4) **MMBT** [16], a multimodal bi-transformer designed to capture both intramodal and intermodal interactions. 5) **VisualBERT COCO** [20], a variant of VisualBERT pretrained on the COCO dataset for

multimodal understanding. 6) **ViLBERT CC** [27], a task-agnostic model that learns joint text-image representations. 7) **MOMENTA** [33], which integrates global and local contextual information from both meme text and image. 8) **MaskPrompt** [3], which constructs masked language modeling prompts by concatenating meme text with generated image captions. 9) **Pro-Cap** [2], an enhanced caption-based variant of MaskPrompt that improves image captioning using PVLMS. 10) **ExplainHM** [21], which employs multimodal debates between LLMs to generate explanations from both harmful and harmless viewpoints, followed by fine-tuning a SLM to serve as a debate judge for harmfulness prediction.

C IMPLEMENTATION DETAILS

C.1 Implementation of CasualHM

All experiments were conducted on a Linux server equipped with an NVIDIA RTX A6000 GPU. When estimating $P(r \mid \text{do}(M))$, we employed LLaVA-1.5-13B [24] for English datasets and llava-onevision-qwen2-7b-si-hf [19] for Chinese datasets to generate reasoning sequences. The diverse beam search was configured with six groups, a diversity penalty of 0.8, and three output sequences ($B = 3$). When estimating $P(Y \mid \text{do}(r))$, CausalHM used BridgeTower ("bridgetower-base") [43] as the multimodal encoder and BERT ("bert-base-uncased") [9] as the textual encoder for English datasets. For Chinese datasets, CausalHM employed Chinese-CLIP ("chinese-clip-vit-base-patch16") [45] as the multimodal encoder and BERT ("chinese-bert-wwm-ext") [8] as the textual encoder. During training, we froze the lower layers of both the multimodal encoder and BERT, and fine-tuned only the top layers. Table 5 summarizes the hyperparameter configurations.

Table 5: Hyper-parameters.

Hyper-Parameter	Harm-C	Harm-P	FHM	TOXIC MM
λ	1.3	1.0	1.05	1.45
Batch size	64	32	64	64
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning rate	1e-4	1e-4	9e-4	7e-4
Warmup ratio	0.01	0.01	0.01	0.04
Warmup strategy	Linear	Linear	Linear	Linear
Weight decay	1e-5	1e-5	1e-5	1e-5

C.2 Prompting GPT-5-mini for Automatic Evaluation of Explainability

Since the GPT-5-mini API supports multimodal inputs (images and text), we include meme images and their embedded text within user prompts. Following [21], we design our system prompt as follows:

"Suppose you have been specially designed to perform an explanation evaluation for the harmful meme detection task, you are required to score the provided explanations given the meme text and image. The image and the textual content in the meme are often uncorrelated, but its overall semantic information is presented holistically. Thus it is important to note that you are prohibited from relying on your own imagination, as your goal is to provide the most accurate and reliable score possible."



Figure 6: More examples of correctly predicted harmful memes from the English dataset (HarmC) and the Chinese dataset (ToxiCN-MM) are presented.

Furthermore, to prompt GPT-5-mini to automatically evaluate explanations for each criterion, we use the following user-prompt template:

"Given a meme containing the text [T] and the label [l_{gold}], please evaluate the following two explanations according to the criterion [Criterion]. For each explanation, assign an integer score values on a rating scale from 1 (worst) to 5 (best). 1) [Explanation_causalhm]; 2) [Explanation_explainhm]. Carefully analyze the explanations before assigning scores. Then, return the result strictly in the following JSON format: {'score1': '<score1>', 'score2': '<score2>'}. "

D MORE CASE STUDY

Here we provide additional two cases to show the advantage of our CausalHM over ExplainHM *w.r.t* explainability. Figure 6(a) depicts a harmful meme targeting Donald Trump during the COVID-19 pandemic. ExplainHM-generated explanation is monothematic — the meme is offensive to pandemic workers. In contrast, CausalHM-generated explanation spans four facets: personal attack on Trump,

references to sensitive historical events, dissemination of misinformation or propaganda, and evocation of strong emotional responses, yielding a richer, more persuasive narrative. Similar phenomenon is observed in Figure 6(b), a Chinese harmful meme. ExplainHM flags only "abusive language", whereas CausalHM enumerates abusive language, social division, and mental health impact. The findings demonstrate our CausalHM outperforms ExplainHM in both detection accuracy and interpretability.

E ERROR ANALYSIS

To elucidate failure causes and guide future improvements, we inspect memes misclassified by CausalHM. We can conclude from this failure case that the model's performance is contingent upon the multimodal reasoning quality of the MLLMs that sample the front-door mediator. As shown in Figure 7, the harmful meme references the "covfe" incident involving the U.S. President Donald Trump. However, low image resolution prevents the MLLM (LLaMA) from recognizing Trump and does not retrieve the requisite background.

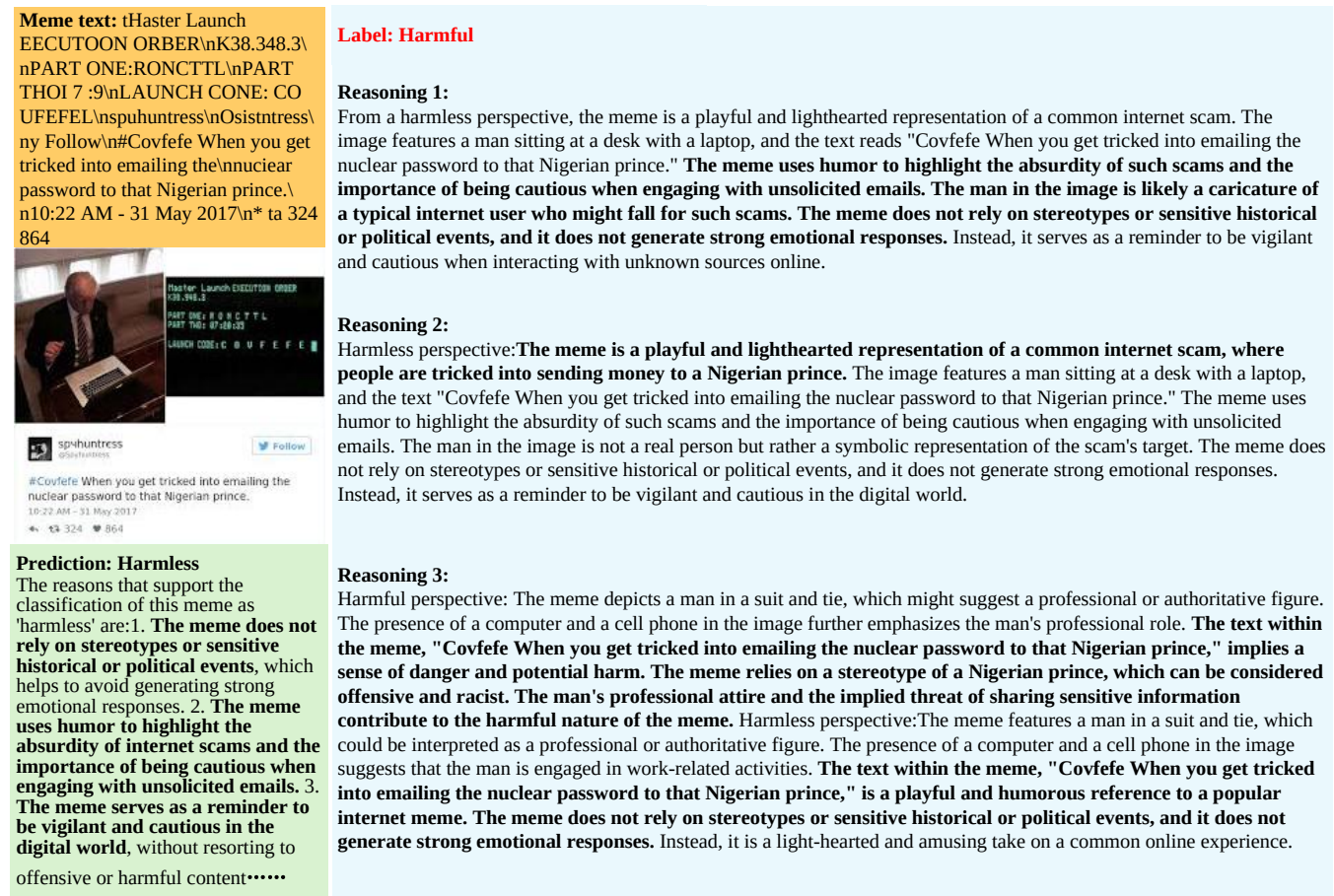


Figure 7: Examples of incorrectly predicted harmful memes from the English dataset (HarmC) are presented.

Consequently, the MLLM tends to generate reasoning from a harmless perspective, and the corresponding harmful arguments lack persuasive force, ultimately leading to a false-negative prediction. In response, we will (i) systematically compare alternative MLLMs as front-door samplers and (ii) augment parametric knowledge with retrieval-augmented generation (RAG) or other complementary strategies to supply the contextual cues.

F LIMITATIONS AND FUTURE WORK

Upon a critical review of the model architecture and a comprehensive empirical evaluation, we identify the following limitations as priority targets for future research:

- 1) Front-door adjustment faces a great challenge in learning the mediator variable. Although we approximate the causal effect of the treatment on the outcome using MLLMs and NWGM, more accurate estimation necessitates new methods in the future.
- 2) Experimental results indicate that the cross-entropy loss treats causal effect estimation as vanilla classification, inviting overfitting to the mediator distribution. Future work will investigate alternative loss functions suitable for causal inference.
- 3) Explanations generated based on reasoning sequences and predictions fail to reflect the debiasing effect on the model. We will

explore a new explanation generation mechanism, rendering the debiasing effect transparent to users.

G ETHICS AND BROADER IMPACT

This paper includes examples of hateful and sarcastic memes that may be disturbing to some readers. These examples are sampled from public datasets (HarmC and TOXIC-MM) solely for illustrative research purposes and do not represent the authors' opinions. CausalHM is designed to diminish exposure to harmful memes, specifically hate speech, harassment, or violence-inciting content, by improving detection accuracy and transparency. Its front-door adjustment module targets getting rid of spurious dataset biases, while reasoning and explanations generated by LLMs serve as human-reviewable aids rather than infallible verdicts.